25% Sample vs. Full-Fleet Equivalence

UrbanEV–Maryland paired-run comparison (final: both runs converged)

UrbanEV–Maryland verification series

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Purpose

All reported results run on a deterministic hash-based 25% sample of the 148,302-owner EV fleet, with charger plug counts scaled by 0.25 (floor one), flow capacity 0.25, storage capacity $0.25^{0.75} \approx 0.35$, and fleet quantities scaled back by four. This note checks that design against a paired full-fleet run of the identical model: same plans (pre-sampling), same charger network at full plug counts, same prices, same AADT-loaded network, same 50-iteration protocol. Both runs are converged (50 iterations); this is the definitive comparison.

Preliminary comparison

Metric	25% × 4	Full fleet	Difference
Charging agents (fleet)	69,652	69,479	−0.2%
Sessions per day (fleet)	45,376	45,435	+0.1%
Home session share	78.1%	77.7%	−0.4 pp
Work session share	13.4%	13.3%	−0.1 pp
Public L2 session share	8.2%	8.2%	−0.0 pp
DCFC + Tesla session share	0.34%	0.24%	−0.10 pp
Home energy share	81.6%	81.6%	0.0 pp
Public energy share	5.7%	5.6%	−0.1 pp
Annual charged energy	374 GWh	372 GWh	−0.5%
ChargePoint occupancy r	0.773	0.811	+5.0%
Session-start r (24 h, weekday)	0.934	0.946	+1.2%
Session-start r (daytime-only)	0.833	0.854	+2.5%

Reading

The sampling design is verified. Fleet totals agree to a fraction of a percent (agents −0.2%, sessions +0.1%, energy −0.5%), the home energy share is identical to one decimal (81.6%), and every venue share matches within 0.4 percentage points. All three timing validations improve at full scale (occupancy r 0.77 to 0.81; 24-hour session starts 0.93 to 0.95; daytime-only 0.83 to 0.85), since four times the agents produce smoother diurnal profiles. The pre-convergence gaps observed mid-run (public L2 +1.6 pp, work −2.0 pp, agents +12.7%) closed entirely by convergence, as predicted from the drift pattern of the 25% baseline.

The one residual difference is DC fast charging: 0.24% of sessions in the full fleet versus 0.34% in the sample (-0.10 pp on a very small base). The plug-count floor (minimum one plug per scaled station) makes DC capacity relatively more abundant in the 25% run; since DCFC is already documented as structurally under-represented (no inter-regional travel), this does not affect any conclusion.

Use in the manuscript

Final sentence for the sampling subsection: "A paired full-fleet run confirms the sample's adequacy: fleet totals agree within 0.5%, the home energy share is identical (81.6%), every venue share matches within 0.4 percentage points, and the observational timing validations improve slightly at full scale (occupancy r 0.77 to 0.81; session starts 0.93 to 0.95)."